

Matas Vitkauskas

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Education

McGill University, Montreal, Canada

Aug 2025 – Present

PhD in Human Genetics

Thesis topic: Single-cell spatial genomics of early life adversity and depression.

Yale-NUS College, Singapore

Aug 2018 – May 2022

Major: Life Sciences; Minor: Physical Sciences (Chemistry)

CAP: 4.47/5.00; Major CAP: 4.79/5.00

Publications

- Xie, J. J. *, **Vitkauskas, M. ***, Do, Q., Saw, T. Y., Sun, A. X., Yang, L., Soong, T. W., Lim, K.-L., Tan, E.-K., Ng, H.-H. & Liu, J. Single-cell transcriptomics revealed molecular vulnerability in a human midbrain-like organoid model of Parkinson's disease. *iScience* 29, (2026). <https://doi.org/10.1016/j.isci.2026.114674>
- **Vitkauskas, M.** & Mathuru, A. S. Total Recall: Lateral Habenula and Psychedelics in the Study of Depression and Comorbid Brain Disorders. *IJMS* 21, 6525 (2020). <https://doi.org/10.3390/ijms21186525>

*These authors contributed equally.

Awards and Honors

- Onni Foundation Award (2000 CAD).
- Lithuania Foundation Award 2026 (3000 USD).
- McGill Graduate Excellence Award (15,580 CAD).
- 2nd place in 10x Genomics Spatial Transcriptomics Hackathon 2024 (prize award 200 SGD).
- Harvard Summer Honors Undergraduate Research Program (SHURP) scholar (grant award 3,000 USD).
- NUS Outstanding Undergraduate Researcher Prize 2020 (cash award 2,000 SGD).
- Yale-NUS Summer Research Program scholar 2019 (grant award 3,500 SGD).

Research Experience

McGill University, PhD Student, Canada

Jul 2025 – Present

- PhD program in Human Genetics. Supervisor: Dr. Gustavo Turecki.
- Generating and analyzing a Visium HD dataset for multi-omic spatial characterization of major depressive disorder in the human prefrontal cortex.
- Analyzing RNA-seq and EM-seq data from prefrontal cortex pyramidal neurons of individuals with depression who died by suicide and had a history of childhood abuse.
- Studying the paraventricular thalamus using single-nucleus RNA-seq in the context of childhood abuse.

Genome Institute of Singapore (GIS), Research Officer, Singapore

Jul 2022 – Jul 2025

- Supervisor: Dr. Jinyue Liu.
- Preparing a first-author paper on temporal and spatial characterization of a human midbrain-like organoid model of cellular vulnerability in Parkinson's disease.

- Analyzed in-house and publicly available scRNAseq and MERFISH spatial transcriptomics data.
- Studied glioblastoma multiforme tumor biopsies and ligand–receptor interactions between invasive tumor subtypes and their immediate cellular neighbourhoods using CosMx spatial transcriptomics.
- Characterized cell-type-specific differential expression effects caused by *Smchd1* mutation in embryonic gonads of mice in collaboration with Dr. Xue Shifeng’s lab at NUS.

Yale-NUS College, Undergraduate Researcher, Singapore

Aug 2021 – May 2022

- Supervisor: Dr. Jan Gruber.
- Wrote bachelor’s thesis titled *Cosmetic neurology at scale: developing a high-throughput phenotypic screening for novel nootropics in D. melanogaster*.
- Created custom Arduino behavioural set-up for vibrational conditioning, CAFÉ feeding assay using capillaries, and a hypoxia recovery assay.

Brigham and Women’s Hospital, Summer Research Intern, Online

May 2021 – Sept 2021

- Supervisor: Dr. William Renthall.
- Studied cell-type-specific enhancers in dorsal root ganglion.
- Performed analysis of scRNAseq and complementary scATACseq datasets in mice.

Institute of Molecular and Cell Biology (IMCB), Intern, Singapore

May 2020 – May 2021

- Supervisor: Dr. Ajay Mathuru.
- Ran behavioural tests investigating the role of oxytocin receptors in scotophobic and thigmotaxic behaviour using OpenCV-based analysis.

Yale-NUS College, Lab Assistant, Singapore

Sept 2019 – Dec 2019

- Supervisor: Dr. Philip Johns.
- Extracted over 100 RNA samples for a study investigating aggressive behaviour in *t. discus* flies.

Duke-NUS Centre for Computational Biology, Intern, Singapore

May 2019 – Jul 2019

- Supervisor: Dr. Steven Rozen.
- Evaluated the performance of software used in mutational signature analysis (e.g. SigProfiler, SignatureAnalyzer, signeR, deconstructSigs).
- Formatted variant calling files (VCFs) for downstream analysis.

Volunteering Experience

Academic Buddy LT, Student Mentor (online)

Aug 2018 – Present

Singapore Society for Neuroscience (SfN.SG), Newsletter Editor and Web Master, Singapore

Mar 2023 – Jul 2025

Integrative Neuroscience Association, Staff Writer, Lithuania

Dec 2020 – Mar 2023

Computer Skills and Languages

Languages: Lithuanian (Native), English (Fluent), French (Intermediate, B1), Chinese (Elementary).

Technical skills: RStudio (Advanced), Python (Intermediate), OCaml (Elementary), C++ (Elementary).

Interests: DJing, classical guitar, basketball, snowboarding.